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0.Setup & Load Data

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns',None)
%matplotlib inline
```

```
In [2]: data={
    "CASENUM": [5,3,66,70,96,97,10,36,51,55],
    "CLMSEX": [0,1,0,1,0,1,0,1,1,1],
    "CLMINSUR": [1.0,0.0,1.0,1.0,1.0,1.0,np.nan,np.nan,1.0,1.0],
    "SEATBELT": [0,0,0,1,0,0,0,0,0,0],
    "CLMAGE": [50.0,18.0,5.0,31.0,30.0,35.0,9.0,34.0,60.0,np.nan],
    "LOSS": [34.940,0.891,0.330,0.037,np.nan,0.309,3.538,4.881,0.882,np.nan],
    "ATTORNEY": [0,1,1,0,1,0,0,0,1,1],
}
df=pd.read_csv("D:\\All Datasets\\claimants sample.csv")if False else pd.DataFrame(
df
```

```
Out[2]:
```

	CASENUM	CLMSEX	CLMINSUR	SEATBELT	CLMAGE	LOSS	ATTORNEY
0	5	0	1.0	0	50.0	34.940	0
1	3	1	0.0	0	18.0	0.891	1
2	66	0	1.0	0	5.0	0.330	1
3	70	1	1.0	1	31.0	0.037	0
4	96	0	1.0	0	30.0	NaN	1
5	97	1	1.0	0	35.0	0.309	0
6	10	0	NaN	0	9.0	3.538	0
7	36	1	NaN	0	34.0	4.881	0
8	51	1	1.0	0	60.0	0.882	1
9	55	1	1.0	0	NaN	NaN	1

1.Data Understanding(Always do this first)

```
In [3]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   CASENUM     10 non-null    int64
1   CLMSEX      10 non-null    int64
2   CLMINSUR    8 non-null     float64
3   SEATBELT    10 non-null    int64
4   CLMAGE      9 non-null     float64
5   LOSS        8 non-null     float64
6   ATTORNEY    10 non-null    int64
dtypes: float64(3), int64(4)
memory usage: 692.0 bytes

```

In [4]: `df.describe()`

Out[4]:

	CASENUM	CLMSEX	CLMINSUR	SEATBELT	CLMAGE	LOSS	ATTORNEY
count	10.000000	10.000000	8.000000	10.000000	9.000000	8.000000	10.000000
mean	48.900000	0.600000	0.875000	0.100000	30.222222	5.726000	0.500000
std	34.999841	0.516398	0.353553	0.316228	17.859016	11.932106	0.527046
min	3.000000	0.000000	0.000000	0.000000	5.000000	0.037000	0.000000
25%	16.500000	0.000000	1.000000	0.000000	18.000000	0.324750	0.000000
50%	53.000000	1.000000	1.000000	0.000000	31.000000	0.886500	0.500000
75%	69.000000	1.000000	1.000000	0.000000	35.000000	3.873750	1.000000
max	97.000000	1.000000	1.000000	1.000000	60.000000	34.940000	1.000000

In [5]: `df.isnull().sum()`

Out[5]:

```

CASENUM    0
CLMSEX     0
CLMINSUR    2
SEATBELT   0
CLMAGE     1
LOSS        2
ATTORNEY    0
dtype: int64

```

2. Missing Value Handling

2.1 Method 1

In [6]: `df1=df.dropna()`
df1

Out[6]:

	CASENUM	CLMSEX	CLMINSUR	SEATBELT	CLMAGE	LOSS	ATTORNEY
0	5	0	1.0	0	50.0	34.940	0
1	3	1	0.0	0	18.0	0.891	1
2	66	0	1.0	0	5.0	0.330	1
3	70	1	1.0	1	31.0	0.037	0
5	97	1	1.0	0	35.0	0.309	0
8	51	1	1.0	0	60.0	0.882	1

In [7]: `df1.isnull().sum()`

Out[7]:

```
CASENUM      0
CLMSEX       0
CLMINSUR     0
SEATBELT     0
CLMAGE       0
LOSS         0
ATTORNEY     0
dtype: int64
```

2.2 Method 2

In [8]:

```
mean=df["CLMAGE"].mean()
median=df["LOSS"].median()
mode=df["CLMINSUR"].mode()[0]

print("Mean CLMAGE",mean)
print("Median LOSS",median)
print("Mode CLMINSUR",mode)
```

Mean CLMAGE 30.22222222222222
 Median LOSS 0.8865000000000001
 Mode CLMINSUR 1.0

In [9]:

```
df["CLMAGE"]=df["CLMAGE"].fillna(mean)
df["LOSS"]=df["LOSS"].fillna(median)
df["CLMINSUR"]=df["CLMINSUR"].fillna(mode)

df.isnull().sum()
```

Out[9]:

```
CASENUM      0
CLMSEX       0
CLMINSUR     0
SEATBELT     0
CLMAGE       0
LOSS         0
ATTORNEY     0
dtype: int64
```

2.3 Method 3

In [10]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
pd.set_option('display.max_columns',None)
%matplotlib inline
```

```
In [11]: from sklearn.impute import SimpleImputer

df_si=pd.DataFrame(data)

mean_imputer=SimpleImputer(strategy='mean')
df_si['LOSS']=mean_imputer.fit_transform(df_si[['LOSS']])

median_imputer=SimpleImputer(strategy='median')
df_si['CLMAGE']=median_imputer.fit_transform(df_si[['CLMAGE']])

mode_imputer=SimpleImputer(strategy='most_frequent')
df_si['CLMINSUR']=mode_imputer.fit_transform(df_si[['CLMINSUR']])

df_si.isnull().sum()
```

```
Out[11]: CASENUM    0
          CLMSEX    0
          CLMINSUR  0
          SEATBELT  0
          CLMAGE    0
          LOSS      0
          ATTORNEY  0
          dtype: int64
```

3.Outlier Detection

```
In [12]: df=pd.DataFrame(data)
          df
```

```
Out[12]:
```

	CASENUM	CLMSEX	CLMINSUR	SEATBELT	CLMAGE	LOSS	ATTORNEY
0	5	0	1.0	0	50.0	34.940	0
1	3	1	0.0	0	18.0	0.891	1
2	66	0	1.0	0	5.0	0.330	1
3	70	1	1.0	1	31.0	0.037	0
4	96	0	1.0	0	30.0	NaN	1
5	97	1	1.0	0	35.0	0.309	0
6	10	0	NaN	0	9.0	3.538	0
7	36	1	NaN	0	34.0	4.881	0
8	51	1	1.0	0	60.0	0.882	1
9	55	1	1.0	0	NaN	NaN	1

3.1 IQR Method

```
In [13]: Q1=df['LOSS'].quantile(0.25)
          Q3=df['LOSS'].quantile(0.75)
```

```
IQR =Q3 - Q1  
  
print("Q1:",Q1)  
print("Q3:",Q3)  
print("IQR:",IQR)
```

Q1: 0.32475

Q3: 3.87375

IQR: 3.549

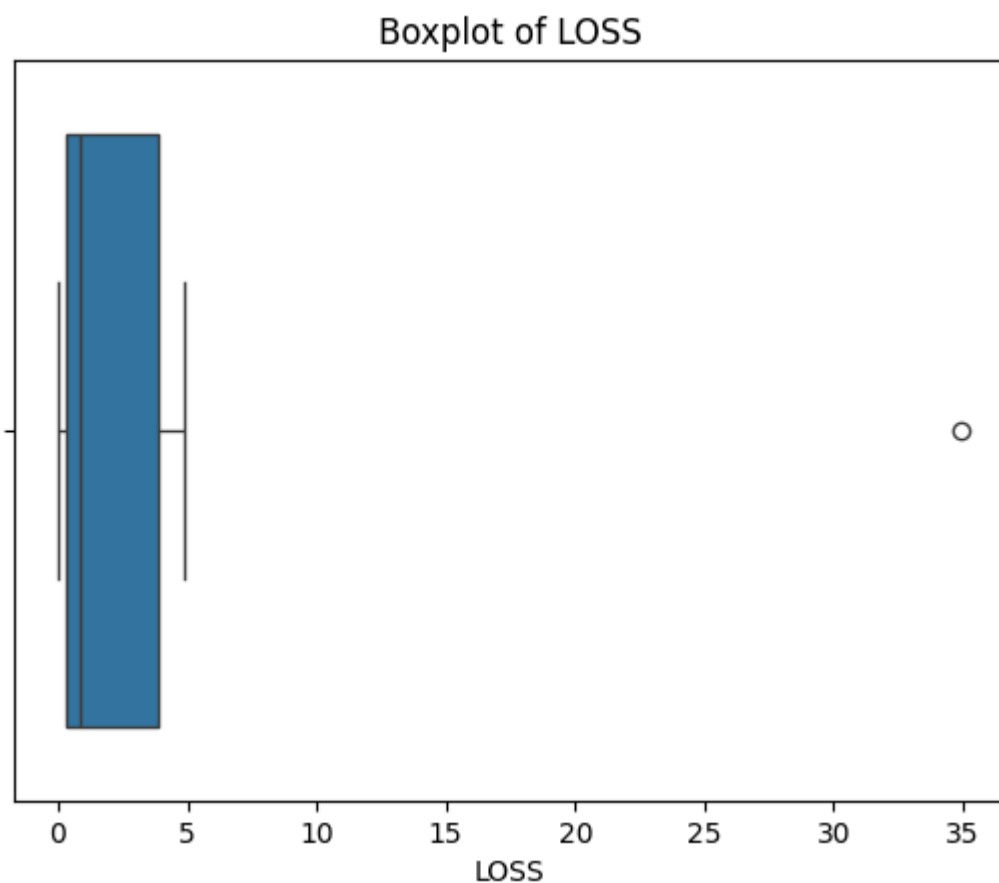
```
In [14]: l_limit=Q1-(1.5*IQR)  
         u_limit=Q3+(1.5*IQR)  
  
         print("Lower limit:",l_limit)  
         print("Upper limit:",u_limit)
```

Lower limit: -4.99875

Upper limit: 9.19725

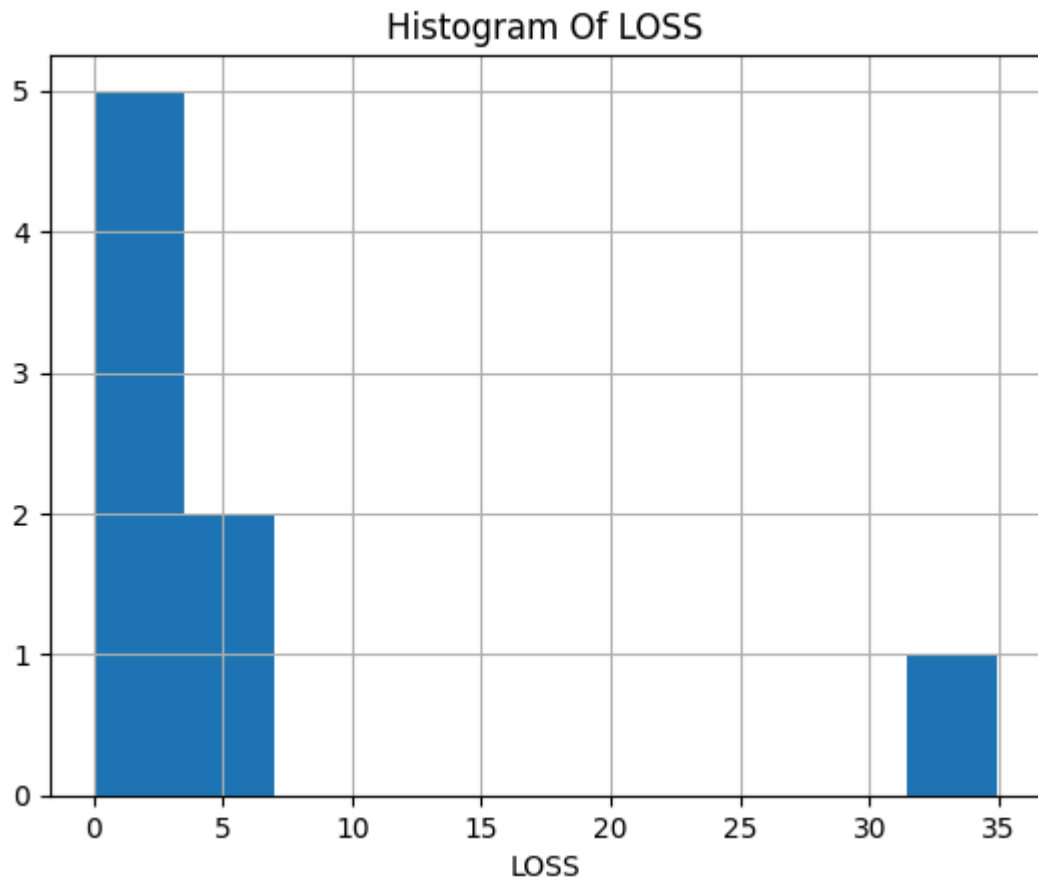
3.2 Boxplot Visualization

```
In [15]: sns.boxplot(x=df['LOSS'])  
         plt.title("Boxplot of LOSS")  
         plt.show()
```

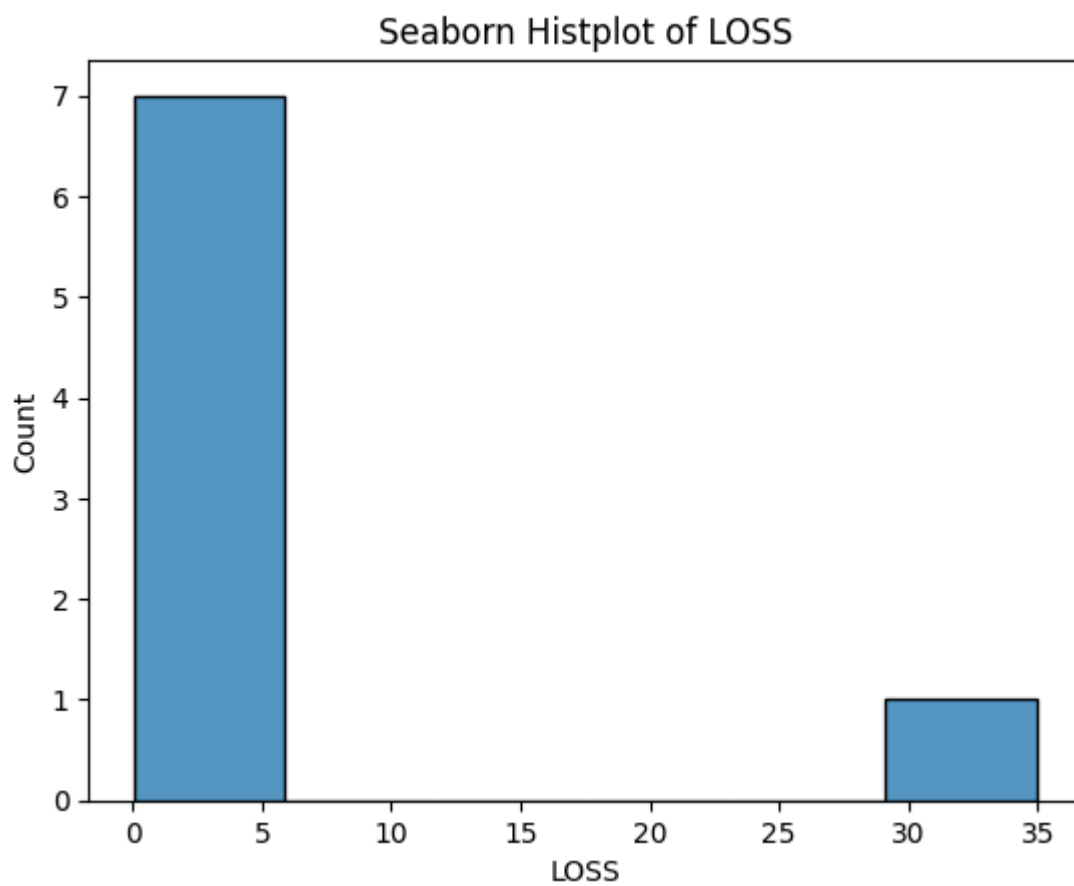


3.3 Histogram Visualization

```
In [16]: df['LOSS'].hist()  
         plt.title("Histogram Of LOSS")  
         plt.xlabel("LOSS")  
         plt.show()
```



```
In [17]: sns.histplot(x=df['LOSS'])  
plt.title("Seaborn Histplot of LOSS")  
plt.show()
```



3.4 Z-score Method

```
In [18]: mu=df['LOSS'].mean()
st=df['LOSS'].std()
print("Mean:",mu,"| Std:",st)
```

Mean: 5.725999999999999 | Std: 11.93210645994363

```
In [19]: df['loss_zscore']=(df['LOSS']-mu)/st
df[['LOSS','loss_zscore']]
```

```
Out[19]:
```

	LOSS	loss_zscore
0	34.940	2.448352
1	0.891	-0.405209
2	0.330	-0.452225
3	0.037	-0.476781
4	NaN	NaN
5	0.309	-0.453985
6	3.538	-0.183371
7	4.881	-0.070817
8	0.882	-0.405964
9	NaN	NaN

```
In [20]: outliers=df[(df['loss_zscore']>2) |(df['loss_zscore']<-2)]
outliers
```

```
Out[20]:
```

	CASENUM	CLMSEX	CLMINSUR	SEATBELT	CLMAGE	LOSS	ATTORNEY	loss_zscore
0	5	0	1.0	0	50.0	34.94	0	2.448352

◀ ————— ▶

```
In [21]: df.drop('loss_zscore',axis=1,inplace=True)
df
```

Out[21]:

	CASENUM	CLMSEX	CLMINSUR	SEATBELT	CLMAGE	LOSS	ATTORNEY
0	5	0	1.0	0	50.0	34.940	0
1	3	1	0.0	0	18.0	0.891	1
2	66	0	1.0	0	5.0	0.330	1
3	70	1	1.0	1	31.0	0.037	0
4	96	0	1.0	0	30.0	NaN	1
5	97	1	1.0	0	35.0	0.309	0
6	10	0	NaN	0	9.0	3.538	0
7	36	1	NaN	0	34.0	4.881	0
8	51	1	1.0	0	60.0	0.882	1
9	55	1	1.0	0	NaN	NaN	1

4.Dealing with Outliers

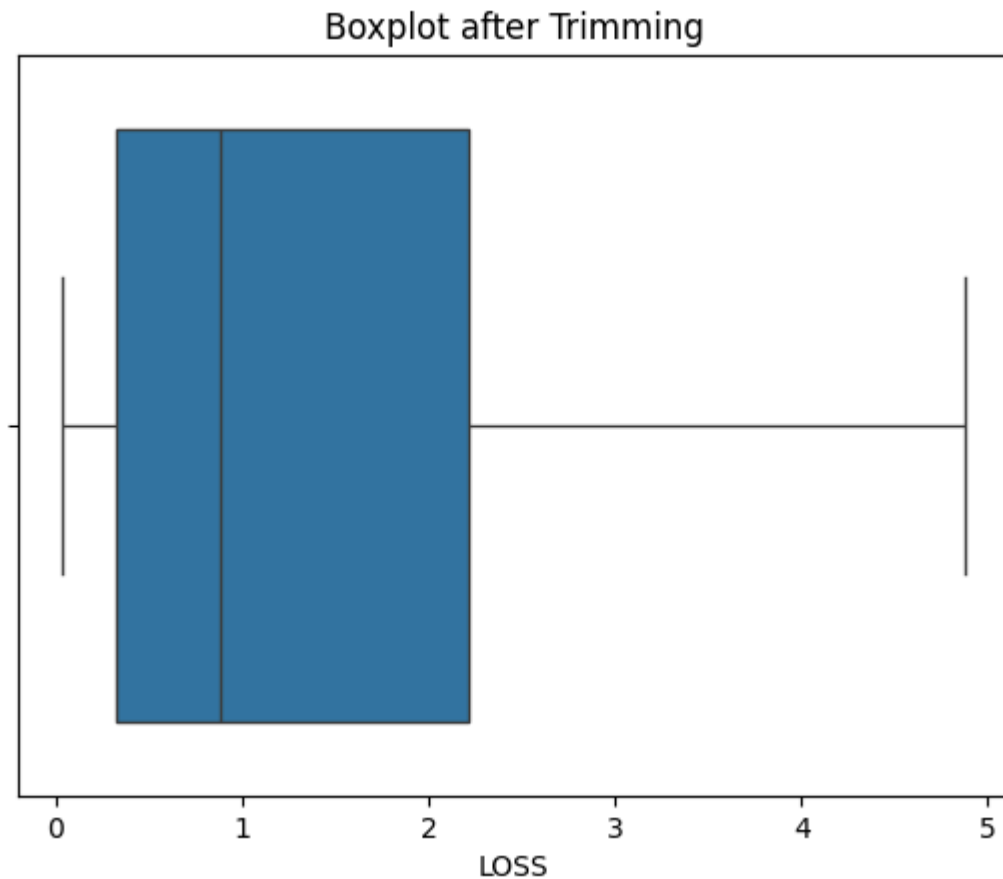
4.1 Trimming (Remove)

In [22]: `df_trimmed=df[(df['LOSS']>l_limit)&(df['LOSS'] < u_limit)]`
`df_trimmed`

Out[22]:

	CASENUM	CLMSEX	CLMINSUR	SEATBELT	CLMAGE	LOSS	ATTORNEY
1	3	1	0.0	0	18.0	0.891	1
2	66	0	1.0	0	5.0	0.330	1
3	70	1	1.0	1	31.0	0.037	0
5	97	1	1.0	0	35.0	0.309	0
6	10	0	NaN	0	9.0	3.538	0
7	36	1	NaN	0	34.0	4.881	0
8	51	1	1.0	0	60.0	0.882	1

In [23]: `sns.boxplot(x=df_trimmed['LOSS'])`
`plt.title("Boxplot after Trimming")`
`plt.show()`



4.2 Winsorization

```
In [24]: df_clean=df.dropna()  
df_clean.isnull().sum()
```

```
Out[24]: CASENUM      0  
CLMSEX      0  
CLMINSUR    0  
SEATBELT    0  
CLMAGE      0  
LOSS        0  
ATTORNEY    0  
dtype: int64
```

```
In [25]: from feature_engine.outliers import Winsorizer  
  
win=Winsorizer(capping_method='iqr',tail='both',fold=1.5,variables=['LOSS'])  
df_win=win.fit_transform(df_clean[['LOSS']])  
df_win
```

Out[25]: **LOSS**

0 1.7505

1 0.8910

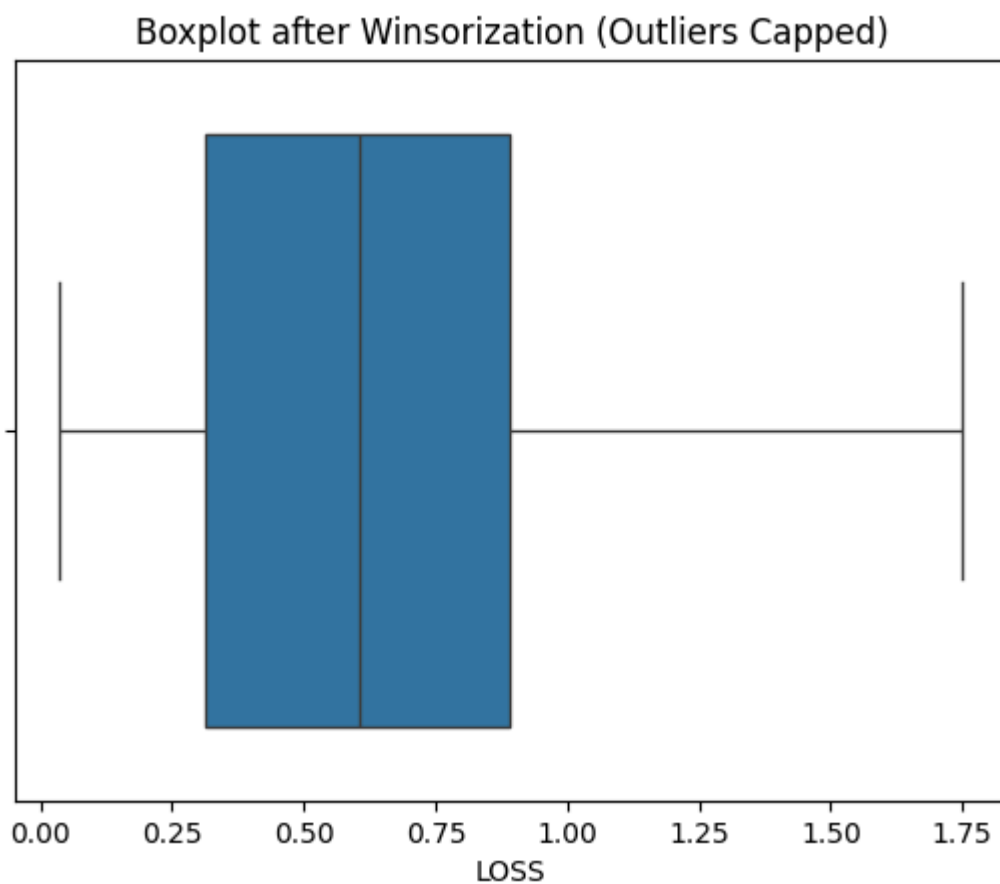
2 0.3300

3 0.0370

5 0.3090

8 0.8820

```
In [27]: sns.boxplot(x=df_win['LOSS'])  
plt.title("Boxplot after Winsorization (Outliers Capped)")  
plt.show()
```



In []: